

Sowing the Seeds of the Future: Optical Fiber Radically Improves the Digital Level of NAU's Campus

播下未来的种子：光纤如何彻底提升南京农业大学的数字化水平

Huawei Case Study · 南京农业大学（NAU）全光校园改造 · 原文：e.huawei.com

阅读说明：每句英文下方紧跟中文译文（Trancy 式上下排）。这是本课程的**选型落地案例**：一张真实的 XGS-PON + Wi-Fi 6 全光校园网。译文由 AI 生成，仅供学习参考；如遇歧义，以英文原文为准。

案例要素：制式 XGS-PON（10G/10G 对称）· 无线 Wi-Fi 6（802.11ax）· 架构 OLT + 无源分光器 SPL + ONU + 面板 AP（两层）· 地点：南农卫岗校区南苑学生宿舍 · 背景：2020 年远程教学流量激增，原交换机网络无法支撑

正文 / Body

Behind all of its achievements, NAU's faculty, staff, and tens of thousands of students are the foundation for its success, with all of them aspiring to the university's motto: integrity, simplicity, diligence, and benevolence.

在所有这些成就背后，南农的教职员工和数万名学生是其成功的基石，他们共同践行着校训：诚朴勤仁。

And another vital support, of course, is NAU's infrastructure, including an upgraded campus network constructed with cutting-edge Information and Communications Technology (ICT).

当然，另一个重要支撑是南农的基础设施，其中包括用前沿信息通信技术（ICT）建设的升级版校园网络。

Accelerating informatization has become a major goal for universities in China, using ICT to improve teaching levels and gain a competitive advantage for future educational development.

加快信息化已成为中国高校的主要目标——用 ICT 提升教学水平，并在未来教育发展中取得竞争优势。

Deploying a safe, reliable, cost-effective, and sustainable campus network was therefore high on NAU's wish-list.

因此，部署一张安全、可靠、经济且可持续的校园网络，是南农的首要诉求之一。

Indeed, in 2019, it signed a strategic cooperation agreement with Huawei.

事实上，2019 年南农就与华为签署了战略合作协议。

Powered by its latest ICT innovations, products, and solutions, Huawei subsequently provided ICT infrastructure and a top-level design to help NAU build a safe and eco-friendly smart campus.

依托最新的 ICT 创新、产品和方案，华为随后提供了 ICT 基础设施与顶层设计，帮助南农建设安全、环保的智慧校园。

But, as 2020 arrived, remote learning rapidly became the new norm and NAU experienced a sharp increase in demand for concurrent online access.

但到了 2020 年，远程教学迅速成为常态，南农的并发在线访问需求急剧上升。

Unsurprisingly, the legacy network was unable to keep up and cope with resulting traffic surges, forcing NAU's hand to upgrade the network — fast.

不出所料，原有网络无法跟上、也难以应对随之而来的流量洪峰，迫使南农必须尽快升级网络。

The structure of the legacy switch-based campus network was also overly complex, making it difficult to maintain and manage.

原有基于交换机的校园网络结构也过于复杂，维护和管理都很困难。

Wi-Fi interference also severely impacted the user experience.

Wi-Fi 干扰也严重影响了用户体验。

Yan Xianglin, the Vice President in charge of NAU's informatization, led a comprehensive evaluation, ultimately choosing to build a future-proofed all-optical network in the Nanyuan student dormitory on the university's Weigang campus.

负责南农信息化工作的副校长闫祥林牵头进行了全面评估，最终决定在卫岗校区南苑学生宿舍建设一张面向未来的全光网络。

With this project, NAU was cast as a pioneer of 10-Gigabit-Capable Symmetric Passive Optical Networks (XGS-PONs), underpinned by Wi-Fi 6 all-optical network architecture.

通过这个项目，南农成为**万兆对称无源光网络（XGS-PON）**的先行者，其底座是 Wi-Fi 6 全光网络架构。

Devices featured in the solution include an Optical Line Terminal (OLT) aggregation device, SPL passive optical splitter, Optical Network Unit (ONU) for multi-service access, and panel Access Points (APs).

方案中的设备包括：光线路终端（OLT）汇聚设备、SPL 无源分光器、用于多业务接入的光网络单元（ONU），以及面板式无线接入点（AP）。

All-optical deployment for the wired network achieves high-bandwidth access and full-service convergence.

有线网络的全光部署实现了高带宽接入与全业务融合。

XGS-PON technology not only provides high bandwidth for symmetric upstream and downstream access, but also simplifies Operations, Administration, and Maintenance (OAM).

XGS-PON 技术不仅为对称的上下行接入提供高带宽，还简化了运维管理（OAM）。

Meanwhile, wireless APs — adopting the 802.11ax (Wi-Fi 6) standard — provide high bandwidth and low latency, with the ability to communicate with multiple devices concurrently, greatly reducing network congestion.

同时，采用 802.11ax（Wi-Fi 6）标准的无线 AP 提供高带宽、低时延，并能与多台设备并发通信，大幅缓解网络拥塞。

Based on XGS-PON technology, Huawei launched its OptiXstar series terminals, optical devices that provide 10 Gbit/s symmetric upstream and downstream bandwidths, addressing traditional issues in terms of bidirectional communication.

基于 XGS-PON 技术，华为推出了 OptiXstar 系列终端，这些光终端提供 **10 Gbit/s 上下行对称**带宽，解决了传统网络在双向通信上的问题。

In addition, featured Wi-Fi 6 capabilities provide low latency to effectively support latency-sensitive applications, such as online course selection and in-school activities.

此外，其 Wi-Fi 6 能力提供低时延，可有效支撑选课、校内活动等时延敏感应用。

Huawei's passive optical Local Area Network (LAN) adopts a simple two-layer architecture, connecting the aggregation equipment room and student dormitories directly through optical fibers.

华为无源光局域网（POL）采用简单的**两层架构**，用光纤直接把汇聚机房与学生宿舍连接起来。

This has laid a solid foundation for the future development of cloud-based networks for NAU.

这为南农未来向云化网络发展奠定了坚实基础。

Copper cables can cover no more than 100 meters, while optical fibers easily reach 20 kilometers, more than enough to cover the entire NAU campus.

铜缆最远只能覆盖 **100 米**，而光纤可轻松达到 **20 公里**，足以覆盖整个南农校园。

In addition, compared with active aggregation switches on copper networks, passive optical splitters reduce active fault points, significantly simplifying and improving Operations and Maintenance (O&M).

此外，与铜缆网络中的有源汇聚交换机相比，无源分光器减少了有源故障点，显著简化和改善了运维（O&M）。

Upgrading a traditional copper network means that cables have to be replaced: this is a laborious, lengthy job.

升级传统铜缆网络意味着必须更换线缆：这是一项费时费力的工程。

Nonetheless, with the arrival of COVID-19, upgrading bandwidth became a matter of extreme urgency and copper networks proved unable to scale up with anything approaching ease.

然而，随着新冠疫情的到来，带宽升级变得极其紧迫，而铜缆网络根本无法轻松扩容。

Optical networks, however, offer almost unlimited bandwidth and are very easy to upgrade.

相比之下，光网络提供近乎无限的带宽，且升级非常容易。

All that's required is for devices at both ends to be upgraded and network evolution is complete, a smooth process that only takes a few days.

只需升级两端设备，网络演进即可完成——整个过程顺畅，仅需几天。

Traditional copper networks rely on numerous power rooms and access devices to function.

传统铜缆网络的运行依赖大量弱电间和接入设备。

This generates plenty of heat and requires air conditioning systems to remain cool, posing fire safety hazards.

这会产生大量热量，需要空调降温，并带来消防安全隐患。

In comparison, an all-optical solution uses passive optical splitters, which minimize the physical hardware footprint as well as mitigating the risk of fire.

相比之下，全光方案使用无源分光器，既最小化了硬件占地，也降低了火灾风险。

关键术语中英对照 / Key Terms

OLT (Optical Line Terminal)	光线路终端，PON 局端设备（有源）
ONU (Optical Network Unit)	光网络单元，用户侧设备（有源）
SPL (passive optical splitter)	无源分光器，ODN 中把光均分为多路（无源、不供电）
XGS-PON	万兆对称无源光网络：上下行各 10 Gbit/s（ITU-T G.9807.1）
two-layer architecture	两层架构：OLT + ODN + ONU，取代核心-汇聚-接入三层
POL (Passive Optical LAN)	无源光局域网，PON 在企业/院区局域网场景的落地形态
AP (Access Point), 802.11ax	无线接入点；802.11ax 即 Wi-Fi 6
full-service convergence	全业务融合：数据、语音、视频等多种业务跑在一张网上
OAM / O&M	运行、管理与维护（Operations, Administration and Maintenance）
future-proofed	面向未来、可长期演进的

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